Ultrasound Obstet Gynecol 2026
Published online in Wiley Online Library (wileyonlinelibrary.com). **DOI:** 10.1002/uog.70320.

Prognostic performance of umbilical artery Doppler-based classification in monochorionic pregnancies complicated by selective fetal growth restriction

S. SORRENTI[1,2#], O. YAGHI[3,4#], S. PRASAD[4], D. MOHAMMED[4] and A. KHALIL[1,3,4]

[1]*Fetal Medicine Unit, Liverpool Women's Hospital, Liverpool, UK;* [2]*Sapienza University of Rome, Rome, Italy;* [3]*Vascular Biology Research Centre, Molecular and Clinical Science Research Institute, City St George's University of London, London, UK;* [4]*Fetal Medicine Unit, Department of Obstetrics and Gynaecology, St George's University Hospitals NHS Foundation Trust, University of London, London, UK*

KEYWORDS: classification; diagnosis; fetal growth restriction; monochorionic diamniotic; outcome; survival; twin

ABSTRACT

Objectives *To evaluate the performance of the umbilical artery (UA) Doppler-based classification system, published by Gratacós* et al., *in stratifying selective fetal growth restriction (sFGR) by severity and predicting adverse outcomes in affected monochorionic diamniotic (MCDA) pregnancies.*

Methods *This was a retrospective cohort study of MCDA pregnancies complicated by sFGR, seen at a single tertiary center (St George's Hospital, London, UK) from January 2000 to July 2024. Included cases were diagnosed with sFGR using the criteria of the recent Delphi consensus. Fetal growth was evaluated using twin-specific charts. A comparison of perinatal outcomes across the different groups of sFGR severity according to the UA Doppler-based classification was performed. Additionally, the discriminatory performance of this classification system and other ultrasound parameters for adverse outcomes was investigated using receiver-operating-characteristics (ROC)-curve analysis.*

Results *Overall, 107 pregnancies complicated by sFGR were included in the analysis. The majority of cases were classified as Type-I sFGR (83/107 (77.6%)) as per the UA Doppler-based classification, while 15.0% (16/107) of cases were Type II and 7.5% (8/107) were Type III. Superimposed twin–twin transfusion syndrome occurred in 6.0% of Type-I sFGR, 18.8% of Type-II sFGR and 37.5% of Type-III sFGR cases (*P = 0.010*). The rate of intact survival of both twins without perinatal complications was found to be significantly higher in Type-I cases (54.2%) than in Type-II cases (18.8%) (*P = 0.013*). No significant differences were found in the rates of double intrauterine fetal demise (IUFD), IUFD of the smaller twin or composite adverse perinatal outcome (CAPO) of either twin. The discriminatory performance of the UA Doppler-based sFGR classification at diagnosis for predicting the occurrence of CAPO in the smaller twin was low (area under the ROC curve (AUC), 0.615 (95% CI, 0.506–0.723)). The combination of severity classification using the UA Doppler-based system with other variables (early diagnosis of sFGR, fetal weight discordance (%) and estimated fetal weight of one twin < 3rd centile) in different models adjusted for gestational age at delivery, laser photocoagulation and classification stage progression showed good discriminatory performance for predicting CAPO in the smaller twin (AUC > 0.9 for all).*

Conclusions *The UA Doppler-based sFGR classification has limited value in predicting adverse outcomes in pregnancies complicated by sFGR. The combination of severity classification using the UA Doppler-based system with other variables could improve the discriminatory performance and therefore its value in clinical practice. Larger studies are needed to investigate the optimal predictive tool to assess severity in monochorionic pregnancies complicated by sFGR. © 2026 The Author(s). Ultrasound in Obstetrics & Gynecology published by John Wiley & Sons Ltd on behalf of International Society of Ultrasound in Obstetrics and Gynecology.*

Correspondence: Prof. A. Khalil, Fetal Medicine Unit, Department of Obstetrics and Gynaecology, St George's University Hospitals NHS Foundation Trust, Blackshaw Road, London SW17 0QT, UK (e-mail: asmakhalil79@googlemail.com)

#S.S. and O.Y. are joint first authors

Accepted: 21 July 2026

INTRODUCTION

Selective fetal growth restriction (sFGR) in monochorionic (MC) pregnancies is a condition characterized by high perinatal morbidity and mortality[1,2]. The most likely cause is unequal sharing of the placenta, which complicates 10–15% of monochorionic diamniotic (MCDA) twin pregnancies[3–7].

The umbilical artery (UA) Doppler-based sFGR classification system published by Gratacós *et al.*[8], which stratifies the risks of sFGR based on the UA Doppler flow of the smaller twin, has been adopted in clinical practice since its publication in 2007. Abnormal UA Doppler flow patterns may be a result of intertwin vascular anastomoses, as reported in the literature[9–18]. The UA Doppler-based classification delineates three types: Type-I sFGR, which is associated with the best perinatal outcomes and is characterized by positive end-diastolic flow (EDF) in the UA of the smaller twin[8]; Type-II sFGR denotes cases with persistent absent or reversed EDF (AREDF) and is associated with the worst outcomes[8,15]; and Type-III sFGR represents a more unpredictable course, marked by intermittent AREDF[8,15].

The classification system does not incorporate the gestational age (GA) at diagnosis or the variation in UA Doppler in the smaller twin during pregnancy[19,20]. The progression of sFGR from Type I to Type II may be a sign of worsening placental insufficiency[19–23]. Additionally, the fluctuating nature of Type-III sFGR further contributes to the unpredictability of Doppler changes and perinatal outcomes[6,24].

Other parameters and complications that are associated with severe adverse perinatal outcomes in MC twins complicated by sFGR are also not incorporated into the classification system. These include abnormal ductus venosus (DV) Doppler, the coexistence of twin–twin transfusion syndrome (TTTS) and intrauterine fetal demise (IUFD) of the smaller twin[6,25–28]. Management options include conservative approaches followed by early delivery, laser photocoagulation or selective termination of the growth-restricted twin to reduce the risk of prematurity-related complications in the cotwin[29,30].

The primary aim of this study was to evaluate the performance of the UA Doppler-based sFGR classification (Type I, Type II or Type III), as assigned at diagnosis, in stratifying the severity of sFGR and predicting adverse perinatal outcomes in MCDA pregnancies.

METHODS

This was a retrospective cohort study that included all MCDA twin pregnancies complicated by sFGR seen at a single tertiary fetal medicine unit in the UK (St George's Hospital, London, UK) from January 2000 to July 2024. As per International Society of Ultrasound for Obstetrics and Gynecology (ISUOG) practice guidelines[30], all pregnancies were dated using the crown–rump length of the larger cotwin in spontaneously conceived twins. Chorionicity was assessed based on the number of placental masses visible on the dating scan, the thickness of the intertwin membrane and the characteristics of the intertwin membrane attachment to the placenta. After 14 weeks' gestation, the head circumference of the larger twin was used to date the pregnancy[30]. MC twin pregnancies were screened and followed up every 2 weeks from 16 weeks until delivery, as indicated in the ISUOG guideline[30]. Estimated fetal weight (EFW) was calculated using fetal biometry parameters[30] and plotted on chorionicity-specific twin charts[31,32]. The intertwin EFW discordance (FWD) was evaluated and recorded. During each ultrasound scan, the amniotic fluid volume in each sac was assessed by measuring the deepest vertical pocket (DVP).

sFGR was defined, using the Delphi diagnostic criteria, as EFW of one twin < 3^{rd} centile for GA, or a combination of at least two of the four following parameters: EFW of one twin < 10^{th} centile; abdominal circumference of one twin < 10^{th} centile; FWD ≥ 25%; and/or UA pulsatility index (UA-PI) of the smaller twin > 95^{th} centile[7]. TTTS was identified by the simultaneous presence of polyhydramnios in one amniotic sac (DVP > 8 cm) and oligohydramnios (DVP < 2 cm) in the other[33].

Fetal Doppler parameters, including the UA-PI (measured in a free loop of the umbilical cord near the fetal abdominal insertion, as per ISUOG guidelines), the middle cerebral artery peak systolic velocity (MCA-PSV) and the DV pulsatility index, were also recorded. Doppler examination of the UA was performed in a free loop of the umbilical cord, with a 0–30° angle of insonation. Twin anemia–polycythemia sequence (TAPS) was diagnosed if the MCA-PSV of the recipient was < 1 multiple of the median (MoM) and the MCA-PSV of the donor was > 1.5 MoM[30] or, based on the new definitions of TAPS proposed by the recent Delphi consensus, if the MCA-PSV of the recipient was < 0.8 MoM and the MCA-PSV of the donor was > 1.5 MoM, or if there was a difference of 1 MoM between the MCA-PSV of the two twins[30].

Pregnancies were identified retrospectively by searching the ultrasound database (ViewPoint version 5.6.26.148; GE Healthcare, Munich, Germany). Individual patient consent was not required owing to the retrospective nature of the study. MC twin pregnancies complicated by TAPS, twin reverse arterial perfusion (TRAP) sequence or structural anomalies were excluded from the study cohort, as were cases diagnosed with TTTS prior to developing sFGR or those presenting with sFGR and TTTS at sFGR diagnosis. Pregnancies with incomplete outcome data were also excluded. When sFGR was diagnosed before 24 weeks' gestation, it was defined as early sFGR.

In the final cohort, perinatal outcomes were assessed and compared between three groups based on the UA Doppler pattern of the smaller twin, which corresponded to sFGR Types I–III according to the UA Doppler-based classification system, as defined in the study of Gratacós *et al.*[8]. Pregnancy outcomes included IUFD, stage progression (from Type I to Type II or III, or from Type III to Type II), selective termination of pregnancy (sTOP), development of TTTS, laser photocoagulation and GA at delivery among live births. The rate of double IUFD or IUFD of the smaller or larger twin was also evaluated.

Neonatal outcomes of both twins included birth weight, birth-weight centile (assessed using Southwest Thames Obstetric Research Collaborative (STORK) twin charts)[34], admission to the neonatal intensive care unit (NICU) and development of respiratory distress syndrome (RDS). Composite adverse perinatal outcome (CAPO) was defined as the occurrence of any of the following perinatal morbidities: 5-min Apgar score ≤ 7, retinopathy of prematurity, severe intraventricular hemorrhage (Stage III or IV), severe periventricular leukomalacia, severe necrotizing enterocolitis, culture-proven sepsis, IUFD or neonatal death before hospital discharge. Intact twin survival was defined as survival without any listed neonatal morbidity or mortality components.

Maternal demographic data were also recorded, including maternal age, ethnicity, body mass index, parity status and mode of conception.

Statistical analysis

Dichotomous variables are presented as *n* (%) while continuous variables are presented as median (interquartile range). The chi-square and Fisher's exact tests were used for the analysis of dichotomous variables, as appropriate. The Shapiro–Wilk test of normality was performed on continuous variables and comparison of continuous variables between the three groups was performed using the Kruskal–Wallis test or ANOVA, depending on the data distribution. When a statistically significant difference was detected, a *post-hoc* analysis was performed to determine which groups were different. To control for Type-I errors, Bonferroni adjustment for multiple comparisons at a variable level was implemented. Therefore, for each comparison, $P < 0.0167$ was considered significant.

In addition, to evaluate the predictive performance of UA Doppler-based sFGR classification at diagnosis, binary logistic regression analysis was performed with CAPO in the smaller twin as the dependent variable and the three-category classification entered as a categorical predictor using indicator (dummy) coding. Type-I sFGR was specified as the reference group. Predicted probabilities derived from the logistic regression model were subsequently used to construct receiver-operating-characteristics (ROC) curves. The area under the curve (AUC) was calculated to assess the overall discriminatory performance of the classification system.

Moreover, UA Doppler-based sFGR classification was evaluated in combination with other clinical variables, and this was achieved using a multivariable logistic regression model that included classification as a categorical predictor and other clinically relevant variables. These variables were FWD, EFW of the smaller twin $< 3^{rd}$ centile, abnormal DV Doppler and early sFGR. The model was also adjusted for GA at delivery, laser photocoagulation and classification stage progression. Reference categories were defined as follows: Type-I sFGR, no classification stage progression, no laser photocoagulation and late sFGR. Variables were incorporated into different models to investigate their prognostic value. Predicted probabilities from the multivariable models were then used to construct the ROC curves of the combined models. To minimize overfitting, the number of predictors included in the model was restricted according to the events-per-variable principle. A two-sided *P*-value < 0.05 was considered statistically significant.

Collinearity of the variables that were selected for inclusion was tested prior to model construction, using tolerance statistics and variance inflation factor. No evidence of significant collinearity among the predictors was observed (Table S1). However, inclusion of DV Doppler data resulted in unstable parameter estimates owing to sparse data and almost complete separation, particularly affecting the laser photocoagulation and the Type-II sFGR subgroups. Therefore, DV Doppler was not included in any of the multivariable models.

In addition, to account for the random effect of twinning, a generalized-estimating-equations (GEE) model with logit link function and exchangeable correlation structure was used to evaluate predictors of CAPO. Odds ratios (ORs) with 95% CIs were reported. GA at delivery, laser photocoagulation, classification stage progression, early sFGR, FWD and sFGR classification at diagnosis were included in the first model. The reference categories were the same as those used in the multivariable model. The second model included the same variables except GA at delivery and laser photocoagulation, which represent the two variables that most influence the outcome as interventions, leaving only indicators of disease severity (UA Doppler-based sFGR classification, classification stage progression, FWD and early sFGR). The analysis was carried out using SPSS version 27 (IBM Corp., Armonk, NY, USA).

RESULTS

Study population

A total of 820 MCDA twin pregnancies were identified through the ultrasound database search. Of these, 713 cases were excluded for the following reasons: missing Doppler pattern details ($n = 3$); uncomplicated MCDA pregnancy ($n = 253$); TTTS at diagnosis ($n = 286$); structural anomaly ($n = 46$); TAPS or TRAP diagnosed before sFGR ($n = 11$); missing outcome data ($n = 88$); IUFD of one cotwin early in pregnancy (before approximately 16 weeks' gestation, prior to screening for monochorionic complications) or miscarriage ($n = 20$); and non-selective FGR ($n = 6$). The remaining 107 pregnancies were complicated by sFGR, as defined by the Delphi diagnostic criteria, and included in the final analysis. The majority ($n = 83$ (77.6%)) of cases were classified as per UA Doppler-based classification as Type-I sFGR, while 16 (15.0%) cases were classified as Type II and eight (7.5%) were classified as Type III. Early sFGR occurred in 75 (70.1%) cases. The maternal characteristics of the included cases are displayed in Table 1. No significant differences were noted between the three groups regarding any of these characteristics, except nulliparity ($P = 0.011$).

Table 1 Maternal characteristics in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies, according to umbilical artery Doppler-based classification[8]

Characteristic	*Type-I sFGR* (n = 83)	*Type-II sFGR* (n = 16)	*Type-III sFGR* (n = 8)	P
Maternal age (years)	32.0 (29.0–36.0)	31.0 (26.2–36.0)	31.5 (28.5–34.0)	0.843
Assisted conception	6/77 (7.8)	4/15 (26.7)	1 (12.5)	0.101
Nulliparous	48/82 (58.5)	13/16 (81.3)	8 (100)	0.011
Maternal BMI (kg/m^2)*	24.4 (21.9–28.9)	23.4 (21.3–28.7)	24.0 (22.7–28.9)	0.829
Ethnicity				0.802
White	56 (67.5)	8 (50.0)	5 (62.5)	
Black	7 (8.4)	1 (6.3)	0 (0)	
Asian	13 (15.7)	4 (25.0)	2 (25.0)	
Mixed	1 (1.2)	0 (0)	0 (0)	
Other	6 (7.2)	3 (18.8)	1 (12.5)	

Data are given as median (interquartile range), *n/N* (%) or *n* (%). *Maternal body mass index (BMI) data were available for 69 Type-I cases, 14 Type-II cases and all eight Type-III cases.

Fetal intervention was performed in 15 cases, including laser photocoagulation in 12 (11.2%) cases, amniodrainage in two (1.9%) cases (both complicated by superimposed TTTS) and intrauterine transfusion in one (0.9%) case (for the larger twin with anemia, diagnosed after IUFD of the smaller twin). Two (1.9%) cases were managed with sTOP of the smaller twin; in one of these, IUFD of the larger twin occurred.

Perinatal outcomes according to umbilical artery Doppler-based classification

A comparison between the three groups based on the UA Doppler-based sFGR classification is displayed in Table 2. The rate of early sFGR was similar, with high prevalence in all groups (66.3% in Type I, 87.5% in Type II and 75.0% in Type III; $P = 0.225$). The rate of sTOP was also similar across the groups ($P = 0.363$). The development of TTTS occurred in 6.0% of Type-I, 18.8% of Type-II and 37.5% of Type-III cases ($P = 0.010$). The classification stage progressed in 14.5% of Type-I cases and 62.5% of Type-III sFGR cases ($P = 0.005$). Laser photocoagulation was performed mostly in Type-II cases (37.5%), followed by Type-III cases (25.0%) and then Type-I cases (4.8%) ($P < 0.001$) (a significant difference was only found between the Type-I and Type-II groups ($P = 0.001$)).

Regarding perinatal outcomes, the rate of intact survival of both twins was significantly higher in the Type-I group (54.2%) compared with the Type-II group (18.8%) ($P = 0.013$), but not compared with the Type-III group (25.0%) ($P = 0.149$). However, these results should be interpreted with caution owing to the high risk of Type-I errors as a result of the relatively small sample sizes of the sFGR Type II and III groups. The rates of double IUFD (25.0%), IUFD of the smaller twin (40.0%) and IUFD of the larger twin (25.0%) in the Type-II sFGR group were found to be higher than observed in the Type-I sFGR group (6.0%, 14.6% and 7.2%, respectively) and the Type-III sFGR group (12.5%, 25.0% and 12.5%, respectively), but there was no significant difference in these parameters between the groups ($P = 0.055$, $P = 0.064$ and $P = 0.098$, respectively). Additionally, no significant differences in GA at delivery of liveborn neonates ($P = 0.131$), rate of CAPO in the smaller twin ($P = 0.019$) or rate of CAPO in the larger twin ($P = 0.037$) were observed between the three groups.

Neonatal outcomes included birth weight and birth-weight centile of the smaller and the larger twin, which were found to be similar across the groups ($P = 0.127$ and $P = 0.110$, respectively, for the smaller twin and $P = 0.580$ and $P = 0.448$, respectively, for the larger twin). Similarly, the rates of NICU admission of both the smaller and larger twin were found to be similar across the groups ($P = 0.057$ and $P = 0.113$, respectively), as were the rates of RDS ($P = 0.072$ for the smaller twin and $P = 0.122$ for the larger twin).

Prediction of adverse outcomes

The discriminatory performance of the UA Doppler-based sFGR classification alone and in combination with other variables as predictors of CAPO in the smaller twin, was assessed. All results of the predictive models are displayed in Tables 3 and 4.

The UA Doppler-based classification at diagnosis alone (Model 1) showed poor discriminatory performance for the considered outcome (AUC, 0.615 (95% CI, 0.506–0.723)) (Table 4, Figure 1). The combination of UA Doppler-based classification with other variables as predictors of CAPO in the smaller twin was evaluated in three different predictive models, each one adjusted for GA at delivery, laser photocoagulation and classification stage progression: Model 2 included early diagnosis of sFGR (AUC, 0.936 (95% CI, 0.890–0.983)); Model 3 included FWD (AUC, 0.938 (95% CI, 0.894–0.983)); and Model 4 included EFW of the smaller twin < 3rd centile (AUC, 0.939 (95% CI, 0.894–0.985)) (Figure 2). Each model showed good discriminatory performance (Tables 3 and 4).

Evaluation of each of the variables included in the models is displayed in Table 3. A direction reversal was observed for the Type-II sFGR group after adjustment, probably reflecting sparse data rather than a true protective effect, given the wide 95% CIs. The overall adjusted models, however, demonstrated good discriminative ability (AUC > 0.9 for all adjusted models) (Table 4).

Table 2 Perinatal outcomes in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies, according to umbilical artery Doppler-based sFGR classification

Parameter	*Type-I sFGR (n = 83)*	*Type-II sFGR (n = 16)*	*Type-III sFGR (n = 8)*	P	P (post-hoc *analysis)**
Early sFGR (< 24 weeks)	55 (66.3)	14 (87.5)	6 (75.0)	0.225	
GA at diagnosis (weeks)	19 (17–26)	17 (16–19)	19 (16.2–23.5)	0.047	
Stage progression†	12/83 (14.5)	0/15 (0)	5/8 (62.5)	< 0.001	0.203†† 0.005‡‡ 0.002§§
Development of TTTS	5 (6.0)	3 (18.8)	3 (37.5)	0.010	0.117†† 0.021‡‡ 0.362§§
Selective termination of pregnancy	1 (1.2)	1 (6.3)	0 (0)	0.363	
Laser photocoagulation	4 (4.8)	6 (37.5)	2 (25.0)	< 0.001	0.001†† 0.085‡‡ 0.667§
Intact survival of both twins	45 (54.2)	3 (18.8)	2 (25.0)	0.015	0.013†† 0.149‡‡ 1.000§§
Double IUFD	5 (6.0)	4 (25.0)	1 (12.5)	0.055	
GA at delivery (among livebirths) (weeks)‡	34 (32–36)	32 (31.2–34.7)	32 (31–35)	0.131	
Outcomes of smaller twin					
Birth weight (g)§	1605.0 (1262.5–1880.0)	1222.5 (893.7–1861.0)	1250.0 (682.5–1905.0)	0.127	
Birth-weight centile§	2.61 (0.65–12.06)	0.19 (0.0–6.06)	0.46 (0.0–20.78)	0.110	
Admission to NICU	16/70 (22.9)	5/9 (55.6)	3/6 (50.0)	0.057	
RDS	18/70 (25.7)	4/9 (44.4)	4/6 (66.7)	0.072	
CAPO¶	32 (38.6)	11 (68.8)	6 (75.0)	0.019	
IUFD**	12/82 (14.6)	6/15 (40.0)	2 (25.0)	0.064	
5-min Apgar score ≤ 7	1/59 (1.7)	1/6 (16.7)	1/5 (20.0)	0.045	
ROP	2/70 (2.9)	1/9 (11.1)	0/6 (0)	0.400	
IVH Stage III/IV	0/70 (0)	0/9 (0)	0/6 (0)	N/A	
Severe PVL	0/70 (0)	1/9 (11.1)	0/6 (0)	0.014	0.005†† 1.000§§
NEC	2/70 (2.9)	0/9 (0)	0/6 (0)	0.803	
Culture-proven sepsis	2/70 (2.9)	0/9 (0)	0/6 (0)	0.803	
Neonatal death before discharge	0/70 (0)	0/9 (0)	1/6 (16.7)	0.001	< 0.001‡‡ 0.400§§
Outcomes of larger twin					
Birth weight (g)§	2039 (1739–2397.5)	1907.5 (1173.5–2516)	1727.5 (1621.2–2297.5)	0.580	
Birth-weight centile§	40.81 (20.67–64.73)	42.72 (5.3–72.9)	51.18 (41.99–69.52)	0.448	
Admission to NICU	18/77 (23.4)	6/12 (50.0)	1/7 (14.3)	0.113	
RDS	25/77 (32.5)	7/12 (58.3)	4/7 (57.1)	0.122	
CAPO¶	31 (37.3)	11(68.8)	5 (62.5)	0.037	
IUFD**	6 (7.2)	4 (25.0)	1 (12.5)	0.098	
5-min Apgar score ≤ 7	1/62 (1.6)	1/8 (12.5)	0/5 (0)	0.722	
ROP	0/77 (0)	1/12 (8.3)	0/7 (0)	0.029	
IVH Stage III/IV	2/77 (2.6)	0/12 (0)	0/7 (0)	0.777	
Severe PVL	0/77 (0)	1/12 (8.3)	0/7 (0)	0.029	
NEC	2/77 (2.6)	1/12 (8.3)	0/7 (0)	0.504	
Culture-proven sepsis	1/77 (1.3)	0/12 (0)	1/7 (14.3)	0.061	
Neonatal death before discharge	0/76 (0)	0/12 (0)	0/7 (0)	N/A	

Data are presented as n (%), median (interquartile range) or n/N (%). In one case of Type-II sFGR, double fetal demise occurred due to selective termination of smaller twin and intrauterine fetal demise (IUFD) of larger twin. **Post-hoc* analysis was performed to compare between each group when a significant difference across the three groups was identified (excluding incidences of $n = 0$ for both groups). †Refers to progression from Type-I to Type-II or -III sFGR, or from Type-III to Type-II sFGR. ‡Gestational age (GA) at delivery was available for 77 cases of Type-I sFGR, 12 cases of Type-II and seven cases of Type-III. §Birth-weight data were available for 70 cases of Type-I sFGR, nine cases of Type-II and five cases of Type-III for the smaller twin and in 76 cases of Type-I sFGR, 12 cases of Type-II and six cases of Type-III for the larger twin. ¶Composite adverse perinatal outcome (CAPO) was defined as the occurrence of at least one of: Apgar score ≤ 7 at 5 min after delivery, retinopathy of prematurity (ROP), severe intraventricular hemorrhage (IVH Stage III/IV), severe periventricular leukomalacia (PVL), severe necrotizing enterocolitis (NEC), culture-proven sepsis, IUFD or neonatal death before hospital discharge. **Among continuing pregnancies that did not opt for selective termination of pregnancy. ††Comparison between Type I and Type II; ‡‡comparison between Type I and Type III; §§ comparison between Type II and Type III. N/A, not available ($n = 0$ in each group); RDS, respiratory distress syndrome; TTTS, twin-to-twin transfusion syndrome.

Multivariable analysis accounting for twinning

Table 5 displays the results of the multivariable GEE analysis for CAPO, adjusted for twin clustering. As in the model for the prediction of CAPO in the smaller twin (Table 3), GA at delivery was strongly and inversely associated with CAPO (OR, 0.483 (95% CI, 0.308–0.759); $P = 0.002$). In this model, laser photocoagulation was also found to be significantly associated with CAPO (OR, 7.034 (95% CI, 1.142–43.317); $P = 0.035$), indicating that this intervention may affect the overall pregnancy outcome. Other

Table 3 Association of predictive variables with composite adverse perinatal outcome in the smaller twin in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies, derived from four multivariable models including different variables

Model	*Included variables*	*OR (95% CI)*	P
Model 1	Type-II sFGR*	3.506 (1.115–11.027)	0.032
	Type-III sFGR*	4.781 (0.909–25.152)	0.065
Model 2	Type-II sFGR*	0.691 (0.090–5.291)	0.722
	Type-III sFGR*	3.616 (0.272–48.039)	0.330
	GA at delivery (in weeks)	0.451 (0.303–0.673)	< 0.001
	Laser photocoagulation	12.279 (0.153–983.02)	0.262
	Stage progression†	1.223 (0.092–16.243)	0.879
	Early sFGR	1.358 (0.393–4.695)	0.629
Model 3	Type-II sFGR*	0.783 (0.101–6.066)	0.815
	Type-III sFGR*	3.600 (0.282–45.953)	0.324
	GA at delivery (in weeks)	0.443 (0.291–0.674)	< 0.001
	Laser photocoagulation	11.606 (0.142–947.50)	0.275
	Stage progression†	1.522 (0.107–21.641)	0.756
	FWD (in %)	0.979 (0.898–1.067)	0.622
Model 4	Type-II sFGR*	0.670 (0.085–5.269)	0.703
	Type-III sFGR*	3.280 (0.256–42.10)	0.362
	GA at delivery (in weeks)	0.461 (0.310–0.684)	< 0.001
	Laser photocoagulation	11.309 (0.179–714.23)	0.251
	Stage progression†	1.258 (0.093–16.945)	0.863
	EFW < 3^{rd} centile	1.262 (0.344–4.624)	0.726

Composite adverse perinatal outcome was defined as the occurrence of at least one of: Apgar score ≤ 7 at 5 min after delivery, retinopathy of prematurity, severe intraventricular hemorrhage, severe periventricular leukomalacia, severe necrotizing enterocolitis, culture-proven sepsis, intrauterine death or neonatal death before hospital discharge. *Classified at the time of diagnosis as per the umbilical artery Doppler-based classification system[8] with Type-I sFGR used as the reference. †Refers to progression from Type-I to Type-II or -III sFGR, or from Type-III to Type-II sFGR. EFW, estimated fetal weight; FWD, fetal weight discordance; GA, gestational age; OR, odds ratio.

Table 4 Discriminatory performance of predictive models including different variables for composite adverse perinatal outcome in the smaller twin in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies

Model	*AUC (95% CI)*	*DR (%)*
Model 1 (UA Doppler-based classification)	0.615 (0.506–0.723)	12.2
Model 2 (UA Doppler-based classification + early sFGR)*	0.936 (0.890–0.983)	67.3
Model 3 (UA Doppler-based classification + FWD)*	0.938 (0.894–0.983)	69.4
Model 4 (UA Doppler-based classification + EFW < 3^{rd} centile)*	0.939 (0.894–0.985)	69.4

Composite adverse perinatal outcome was defined as the occurrence of at least one of: 5-min Apgar score ≤ 7, retinopathy of prematurity, severe intraventricular hemorrhage, severe periventricular leukomalacia, severe necrotizing enterocolitis, culture-proven sepsis, intrauterine death or neonatal death before hospital discharge. False-positive rate was set at < 5% for the evaluation. *Adjusted for gestational age at delivery, laser photocoagulation and progression of umbilical artery (UA) Doppler-based sFGR classification stage (Type I to Type II or III, or Type III to II) during pregnancy. AUC, area under receiver-operating-characteristics curve; DR, detection rate; EFW, estimated fetal weight; FWD, fetal weight discordance.

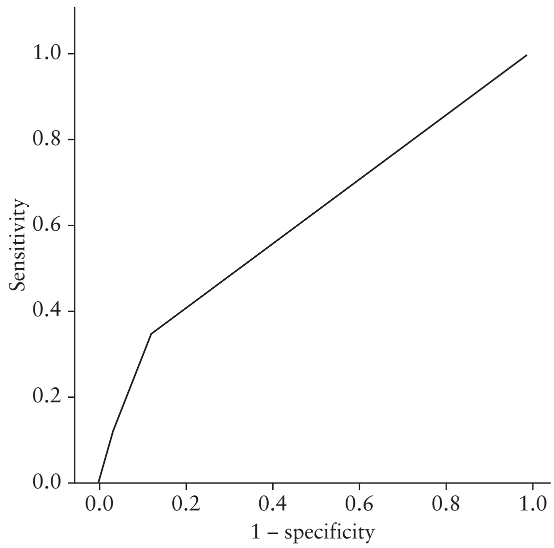

Figure 1 Receiver-operating-characteristics curve for a predictive model for composite adverse perinatal outcome in the smaller twin based on umbilical artery Doppler-based classification of selective fetal growth restriction[8] in 107 affected monochorionic diamniotic twin pregnancies (area under the curve, 0.615 (95% CI, 0.506–0.723)).

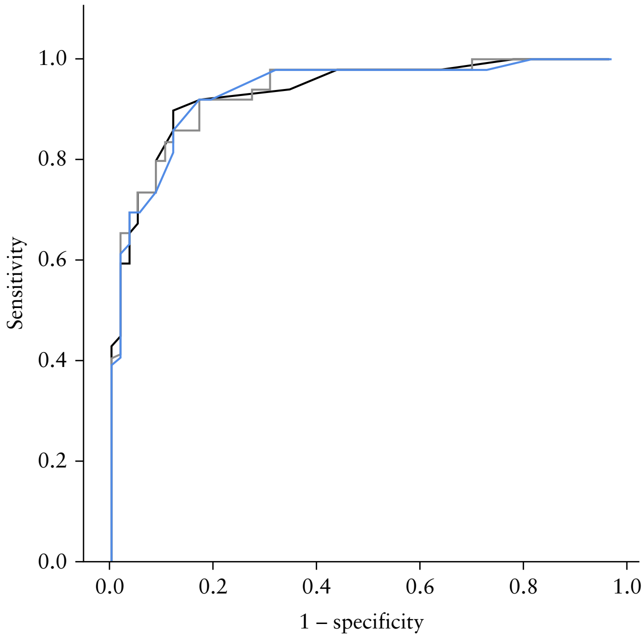

Figure 2 Receiver-operating-characteristics curves for three predictive models for composite adverse perinatal outcome in the smaller twin in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies, adjusted for gestational age at delivery, laser photocoagulation and stage progression (Type-I to Type-II or -III sFGR, or Type-III to Type-II sFGR, as per umbilical artery (UA) Doppler-based sFGR classification[8]) during pregnancy. Model 2 (——) includes UA Doppler-based classification and early sFGR diagnosis (< 24 weeks) (area under the curve (AUC), 0.936 (95% CI, 0.890–0.983)); Model 3 (——) includes UA Doppler-based classification and fetal weight discordance (AUC, 0.938 (95% CI, 0.894–0.983)); and Model 4 (——) includes UA Doppler-based classification and estimated fetal weight of the smaller twin < 3rd centile (AUC, 0.939 (95% CI, 0.894–0.985)).

indicators, including the UA Doppler-based classification, were found to be non-significantly associated, as in the previously described models.

Conversely, the model that excluded GA at delivery and laser photocoagulation, therefore only accounting for disease severity indicators, showed that stage progression, Type-II sFGR and FWD were significant predictors of CAPO for the pregnancy overall (Table 5).

DISCUSSION

Summary of key findings

This study showed that, in terms of intact survival of both cotwins, cases classified as Type-I sFGR as per the UA Doppler-based classification system had significantly better outcomes compared with those defined as Type-II sFGR. The rate of disease progression based on this classification was found to be significantly higher in Type-III sFGR compared with Type-I sFGR cases, with the former also being associated with higher rates of superimposed TTTS. However, other perinatal outcomes considered in our analysis showed no significant differences across the three study groups. The performance of the UA Doppler-based sFGR classification as a predictor of CAPO in the smaller twin was poor. When the classification was combined with other variables, such as FWD or early diagnosis of sFGR, and adjusted in a predictive model, the discriminatory performance improved. However, when adjusted for other variables including stage progression, GA at delivery and laser photocoagulation, the UA Doppler-based classification was found to be non-significantly associated with the adverse outcome.

Table 5 Association of predictive variables with composite adverse perinatal outcome in 107 cases of selective fetal growth restriction (sFGR) in monochorionic diamniotic twin pregnancies derived using two multivariable generalized estimating equations models, accounting for clustering within twin pairs

Model	*Included variables*	*OR (95% CI)*	*P*
Model 1	Type-II sFGR†	0.668 (0.090–4.957)	0.693
	Type-III sFGR†	1.957 (0.398–9.620)	0.409
	GA at delivery (in weeks)	0.483 (0.308–0.759)	0.002
	Laser photocoagulation	7.034 (1.142–43.317)	0.035
	Stage progression‡	1.001 (0.120–8.335)	0.999
	FWD (in %)	1.004 (0.950–1.061)	0.883
	Early sFGR	0.636 (0.244–1.659)	0.355
Model 2*	Type-II sFGR†	4.003 (1.335–12.005)	0.013
	Type-III sFGR†	0.975 (0.148–6.427)	0.979
	Stage progression‡	14.742 (3.293–65.994)	< 0.001
	FWD (in %)	1.063 (1.014–1.114)	0.011
	Early sFGR	1.639 (0.652–4.123)	0.293

Composite adverse perinatal outcome was defined as the occurrence of at least one of: Apgar score ≤ 7 at 5 min after delivery, retinopathy of prematurity, severe intraventricular hemorrhage, severe periventricular leukomalacia, severe necrotizing enterocolitis, culture-proven sepsis, intrauterine death or neonatal death before hospital discharge. *Model 2 was a disease-severity model excluding gestational age (GA) at delivery and laser photocoagulation. †Classification at the time of diagnosis based on umbilical artery Doppler[8] with Type-I sFGR used as the reference. ‡Refers to progression from Type-I to Type-II or -III sFGR, or from Type-III to Type-II sFGR. FWD, fetal weight discordance; OR, odds ratio.

Similar results were found in the multivariable analysis accounting for clustering within twin pairs, with a significant role of GA at delivery in the prediction of CAPO, as well as a strong effect of laser photocoagulation in the occurrence of CAPO in the pregnancy overall.

Interpretation of findings and comparison with existing literature

Classification of sFGR is essential for the correct management of these complex pregnancies. The counseling, surveillance and timing of delivery differ according to the severity of the condition[8]. In the study of Gratacós *et al.*[8], more cases (65/134 (48.5%)) were Type-III sFGR than Type I (39/134 (29.1%)) or Type II (30/134 (22.4%)). In our cohort, most cases (83/107 (77.6%)) were Type-I sFGR, and only 16 (15.0%) and eight (7.5%) cases were Type-II and Type-III sFGR, respectively. This discrepancy could be attributed to the different diagnostic criteria used in the Gratacós cohort for sFGR (EFW < 10^{th} centile in one fetus), and possibly to the improvement in the detection rate of cases with restricted fetal weight and abnormal Doppler data over the years. Moreover, the rate of IUFD of the smaller twin in our cohort was higher than in the Gratacós cohort (Type I, 14.6% *vs* 2.6%; Type II, 40.0% *vs* 0%; Type III, 25.0% *vs* 15.4%). The same observations were made for IUFD of the larger twin. The higher incidence of IUFD in our cohort may be attributed to the fact that, in the Gratacós cohort, just over 38% (13/34) of fetuses with abnormal DV Doppler data, which would presumably put them at risk of further deterioration[35,36] and IUFD, underwent selective cord occlusion, compared with only two fetuses in our cohort undergoing sTOP.

A recent meta-analysis[37] studied the different outcomes of sFGR in MC pregnancies according to the UA Doppler-based classification and showed that the rate of double IUFD was 1.9% (95% CI, 0.5–4.3%) for Type I, 7.0% (95% CI, 3.1–12.5%) for Type II and 4.9% (95% CI, 1.8–9.4%) for Type-III sFGR cases, with a significant difference found only between Type I and Type II (OR, 4.8 (95% CI, 1.3–17.4))[37]. In our cohort, we failed to demonstrate a significant difference in the rate of double IUFD between the three groups, although there was a higher rate in Type-II cases.

Clinical and research implications

In singleton pregnancies complicated by FGR, UA Doppler plays a crucial role in surveillance and management[38]. The most recent ISUOG guideline on ultrasound in twin pregnancy advises using the same recommendations in dichorionic pregnancies complicated by sFGR as in singleton pregnancies[30]. However, sFGR has different implications in MC pregnancies, as Doppler abnormalities may be attributed to the effects of a combination of unbalanced intertwin placental anastomoses and placental insufficiency[16,39], and the management of these cases using laser coagulation is controversial.

In our cohort, almost two-thirds of the Type-III cases progressed to Type II. This has important implications for surveillance: progression to Type II would require prompt intervention, particularly at a viable GA. A previous study has reported the persistence of reversed EDF as one of the criteria for determining the timing of delivery at a viable GA[8]. Type-III sFGR represents a challenge in clinical practice, as previous studies have highlighted the unpredictability of this condition. The study of Chon *et al.*[40] investigated the outcomes of a cohort of Type-III sFGR cases and reported progression into Type II or TTTS in 54% of cases, with a reduced rate of double survival in the cases that progressed (69.2%) compared with the cases that had stable Type-III UA Doppler (95.4%) ($P = 0.0276$). Another multicenter retrospective study[24] aimed to identify predictors of adverse outcomes in Type-III sFGR, and showed a significant association between IUFD and early diagnosis, deterioration of UA Doppler, oligohydramnios in the smaller twin and a large degree of EFW discordance on univariable analysis; however, significance was not maintained in the multivariable analysis. The authors concluded that early diagnosis and UA Doppler deterioration may help to identify cases at higher risk of IUFD.

The UA Doppler pattern is a crucial determinant of fetal outcomes in cases of sFGR, but other clinical characteristics might improve the identification of cases at higher risk of complications, guiding ultrasound surveillance and management. In fact, our data show that the discriminatory performance of this parameter alone is only low, indicating that additional research may be required. As demonstrated by the multivariable predictive model, which included other variables such as GA at delivery, stage progression, laser photocoagulation and early diagnosis, the UA Doppler-based sFGR classification was not significantly associated with the adverse outcome. This suggests that this classification of sFGR, as currently used in clinical practice, may not be sufficient for risk stratification in these pregnancies. A recent study[41] investigated the possible predictors of adverse outcomes in cases of sFGR managed expectantly without considering UA-Doppler pattern and identified oligohydramnios, lack of cycling bladder, DV abnormalities and high MCA-PSV in the smaller twin as potential indicators of adverse perinatal outcomes.

Our results show that the combination of the UA Doppler-based classification with other parameters such as FWD, early diagnosis and EFW of the smaller twin < 3^{rd} centile, could improve the prediction of adverse outcomes in cases of sFGR. Therefore, the inclusion of such indicators in the classification of severity could potentially improve the prognostic performance, and guide counseling and clinical management. As expected, the inclusion of GA at delivery in the adjusted model largely improved the discriminatory performance and, in the multivariable model for CAPO in the smaller twin and the multivariable GEE model accounting for clustering within twin pairs, this remained the only variable significantly associated with the outcome. This aligns with the fact that prematurity represents the most important determinant of perinatal morbidity and mortality. Nevertheless, the identification of additional antenatal predictors of adverse outcome represents a clinical and research challenge.

The role of laser photocoagulation as a predictor of adverse outcome was also investigated: regarding CAPO in the smaller twin, no significant influence was found for this intervention, whereas for the occurrence of CAPO in the pregnancy overall, laser photocoagulation showed a strong effect of association. This is likely owing to the disease severity implied in cases undergoing fetal intervention, as well as the fact that this technique may, in some cases, influence the survival or GA at delivery of the larger twin.

Importantly, over the past two decades, the UA Doppler-based sFGR classification system has proven to be an invaluable tool for guiding immediate clinical decision-making and day-to-day monitoring in MC twin pregnancies. Our data, while limited by sample size and some underpowered analyses, suggest that to improve the prediction of longer-term outcomes, such as adverse perinatal events, there may be value in complementing the current classification system with additional parameters to better address prognostic questions that extend beyond acute management.

Strengths and limitations

To our knowledge, this is the first study investigating the prognostic value of the UA Doppler-based sFGR classification system in clinical practice. This evaluation may help to identify the strengths and pitfalls of this classification system, which might be relevant for future research and clinical practice. Two of the main limitations, mainly due to the nature of the disease, are the relatively small number of cases included in the cohort, especially Type-II and Type-III sFGR cases, and the retrospective design. Moreover, the small number of events, owing to the relatively small sample size, represents a limitation in the construction of predictive models with many variables because of the risk of model overfitting and the risk of instability of the model. Larger studies are necessary to test the model in a large cohort of sFGR fetuses. The lack of long-term follow-up of these cases and the extended study period may also be considered a limitation, as different management strategies may have been adopted because of evolving evidence.

Conclusions

The classification of severity based on the UA Doppler of the smaller twin only partially identifies cases at risk of adverse perinatal outcomes and, therefore, its value in clinical practice is limited. Larger prospective studies are needed to evaluate the ultrasound and clinical parameters, possibly including the GA at sFGR onset, FWD and other variables, that could be predictive of adverse outcomes and therefore guide management of MC twin pregnancies complicated by sFGR.

REFERENCES

1. Simchen MJ, Okrent Smolar AL, Dulitzky M, Sivan E, Morag I. Neonatal morbidities and need for intervention in twins and singletons born at 34–35 weeks of gestation. *J Perinat Med*. 2016;44(8):887-892.
2. Machin GA. The monochorionic twin placenta in vivo is not a black box. *Ultrasound Obstet Gynecol*. 2001;17(1):4-6.
3. Hack KEA, Derks JB, Elias SG, et al. Increased perinatal mortality and morbidity in monochorionic versus dichorionic twin pregnancies: clinical implications of a large Dutch cohort study. *BJOG*. 2008;115(1):58-67.
4. Sebire NJ, Snijders RJ, Hughes K, Sepulveda W, Nicolaides KH. The hidden mortality of monochorionic twin pregnancies. *Br J Obstet Gynaecol*. 1997;104(10):1203-1207.
5. Lewi L, Jani J, Blickstein I, et al. The outcome of monochorionic diamniotic twin gestations in the era of invasive fetal therapy: a prospective cohort study. *Am J Obstet Gynecol*. 2008;199(5):514.e1-514.e8.
6. Khalil A, Thilaganathan B. Selective fetal growth restriction in monochorionic twin pregnancy: a dilemma for clinicians and a challenge for researchers. *Ultrasound Obstet Gynecol*. 2019;53(1):23-25.
7. Khalil A, Beune I, Hecher K, et al. Consensus definition and essential reporting parameters of selective fetal growth restriction in twin pregnancy: a Delphi procedure. *Ultrasound Obstet Gynecol*. 2019;53(1):47-54.
8. Gratacós E, Lewi L, Muñoz B, et al. A classification system for selective intrauterine growth restriction in monochorionic pregnancies according to umbilical artery Doppler flow in the smaller twin. *Ultrasound Obstet Gynecol*. 2007;30:28-34.
9. Gratacós E, Van Schoubroeck D, Carreras E, et al. Impact of laser coagulation in severe twin-twin transfusion syndrome on fetal Doppler indices and venous blood flow volume. *Ultrasound Obstet Gynecol*. 2002;20(2):125-130.
10. Martínez JM, Bermúdez C, Becerra C, López J, Morales WJ, Quintero RA. The role of Doppler studies in predicting individual intrauterine fetal demise after laser therapy for twin-twin transfusion syndrome. *Ultrasound Obstet Gynecol*. 2003;22(3):246-251.
11. Taylor MJO, Shalev E, Tanawattanacharoen S, et al. Ultrasound-guided umbilical cord occlusion using bipolar diathermy for Stage III/IV twin-twin transfusion syndrome. *Prenat Diagn*. 2002;22(1):70-76.
12. Gratacós E, Carreras E, Becker J, et al. Prevalence of neurological damage in monochorionic twins with selective intrauterine growth restriction and intermittent absent or reversed end-diastolic umbilical artery flow. *Ultrasound Obstet Gynecol*. 2004;24(2):159-163.
13. Vanderheyden TM, Fichera A, Pasquini L, et al. Increased latency of absent end-diastolic flow in the umbilical artery of monochorionic twin fetuses. *Ultrasound Obstet Gynecol*. 2005;26(1):44-49.
14. Hecher K, Jauniaux E, Campbell S, Deane C, Nicolaides K. Artery-to-artery anastomosis in monochorionic twins. *Am J Obstet Gynecol*. 1994;171(2):570-572.
15. Wee LY, Taylor MJ, Vanderheyden T, Talbert D, Fisk NM. Transmitted arterio-arterial anastomosis waveforms causing cyclically intermittent absent/reversed end-diastolic umbilical artery flow in monochorionic twins. *Placenta*. 2003;24(7):772-778.
16. Gratacós E, Lewi L, Carreras E, et al. Incidence and characteristics of umbilical artery intermittent absent and/or reversed end-diastolic flow in complicated and uncomplicated monochorionic twin pregnancies. *Ultrasound Obstet Gynecology*. 2004;23(5):456-460.
17. Gaziano E, Gaziano C, Brandt D. Doppler velocimetry determined redistribution of fetal blood flow: correlation with growth restriction in diamniotic monochorionic and dizygotic twins. *Am J Obstet Gynecol*. 1998;178(6):1359-1367.
18. Townsend R, D'Antonio F, Sileo FG, Kumbay H, Thilaganathan B, Khalil A. Perinatal outcome of monochorionic twin pregnancy complicated by selective fetal growth restriction according to management: systematic review and meta-analysis. *Ultrasound Obstet Gynecol*. 2019;53(1):36-46.
19. Khalil A, Liu B. Controversies in the management of twin pregnancy. *Ultrasound Obstet Gynecol*. 2021;57(6):888-902.
20. Mazer Zumaeta A, Gil MM, Rodríguez-Fernández M, et al. Selective fetal growth restriction in monochorionic diamniotic twins: diagnosis and management. *Matern-Fetal Med*. 2022;4(4):268-275.
21. Rustico MA, Consonni D, Lanna M, et al. Selective intrauterine growth restriction in monochorionic twins: changing patterns in umbilical artery Doppler flow and outcomes. *Ultrasound Obstet Gynecol*. 2017;49(3):387-393.
22. Batsry L, Matatyahu N, Avnet H, et al. Perinatal outcome of monochorionic diamniotic twin pregnancy complicated by selective intrauterine growth restriction according to umbilical artery Doppler flow pattern: single-center study using strict fetal surveillance protocol. *Ultrasound Obstet Gynecol*. 2021;57(5):748-755.
23. Colmant C, Lapillonne A, Stirnemann J, et al. Impact of different prenatal management strategies in short- and long-term outcomes in monochorionic twin pregnancies with selective intrauterine growth restriction and abnormal flow velocity waveforms in the umbilical artery Doppler: a retrospective observational study of 108 cases. *BJOG*. 2021;128(2):401-409.
24. Van Mieghem T, Lewi L, Slaghekke F, et al. Prediction of fetal death in monochorionic twin pregnancies complicated by Type-III selective fetal growth restriction. *Ultrasound Obstet Gynecol*. 2022;59(6):756-762.
25. El Emrani S, Groene SG, Verweij EJ, et al. Gestational age at birth and outcome in monochorionic twins with different types of selective fetal growth restriction: a systematic literature review. *Prenat Diagn*. 2022;42(9):1094-1110.
26. Fratelli N, Amighetti S, Bhide A, et al. Ductus venosus Doppler waveform pattern in fetuses with early growth restriction. *Acta Obstet Gynecol Scand*. 2020;99(5):608-614.
27. Peeva G, Bower S, Orosz L, Chaveeva P, Akolekar R, Nicolaides KH. Endoscopic placental laser coagulation in monochorionic diamniotic twins with type II selective fetal growth restriction. *Fetal Diagn Ther*. 2015;38(2):86-93.

28. Monaghan C, Kalafat E, Binder J, Thilaganathan B, Khalil A. Prediction of adverse pregnancy outcome in monochorionic diamniotic twin pregnancy complicated by selective fetal growth restriction. *Ultrasound Obstet Gynecol*. 2019;53(2):200-207.
29. Chalouhi GE, Marangoni MA, Quibel T, et al. Active management of selective intrauterine growth restriction with abnormal Doppler in monochorionic diamniotic twin pregnancies diagnosed in the second trimester of pregnancy. *Prenat Diagn*. 2013;33(2):109-115.
30. Khalil A, Sotiriadis A, Baschat A, et al. ISUOG Practice Guidelines (updated): role of ultrasound in twin pregnancy. *Ultrasound Obstet Gynecol*. 2025;65(2): 253-276.
31. Giorgione V, Briffa C, Di Fabrizio C, Bhate R, Khalil A. Perinatal outcomes of small for gestational age in twin pregnancies: twin vs. singleton charts. *J Clin Med*. 2021;10(4):643.
32. Wright A, Wright D, Chaveeva P, et al. Fetal Medicine Foundation charts for fetal growth in twins. *Ultrasound Obstet Gynecol*. 2024;63(2):181-188.
33. Quintero R, Kontopoulos EV, Barness E, et al. Twin-twin transfusion syndrome in a dichorionic-monozygotic twin pregnancy: the end of a paradigm? *Fetal Pediatr Pathol*. 2010;29(2):81-88.
34. Stirrup OT, Khalil A, D'Antonio F, Thilaganathan B, Southwest Thames Obstetric Research Collaborative (STORK). Fetal growth reference ranges in twin pregnancy: analysis of the Southwest Thames Obstetric Research Collaborative (STORK) multiple pregnancy cohort. *Ultrasound Obstet Gynecol*. 2015;45(3): 301-307.
35. Chmait RH, Chon AH, Korst LM, Stephen Y, Llanes A, Ouzounian JG. Selective intrauterine growth restriction (SIUGR) type II: proposed subclassification to guide surgical management. *J Matern Fetal Neonatal Med*. 2022;35(6): 1184-1191.
36. Nassr AA, Hessami K, Corroenne R, et al. Outcome of laser photocoagulation in monochorionic diamniotic twin pregnancy complicated by Type-II selective fetal growth restriction. *Ultrasound Obstet Gynecol*. 2023;62(3):369-373.
37. Buca D, Pagani G, Rizzo G, et al. Outcome of monochorionic twin pregnancy with selective intrauterine growth restriction according to umbilical artery Doppler flow pattern of smaller twin: systematic review and meta-analysis. *Ultrasound Obstet Gynecol*. 2017;50(5):559-568.
38. Lees CC, Marlow N, van Wassenaer-Leemhuis A, et al. 2 year neurodevelopmental and intermediate perinatal outcomes in infants with very preterm fetal growth restriction (TRUFFLE): a randomised trial. *Lancet*. 2015;385(9983):2162-2172. Erratum in: Lancet. 2015 May 30;385(9983):2152.
39. Gratacos E, Antolin E, Lewi L, et al. Monochorionic twins with selective intrauterinegrowth restriction and intermittent absent or reversed end-diastolic flow (Type III): feasibility and perinatal outcome of fetoscopic placental laser coagulation. *UltrasoundObstet Gynecol*. 2008;31:669-675.
40. Chon AH, Ma SY, Korst LM, Chmait HR, Purnell ME, Chmait RH. Antenatal course of referred monochorionic diamniotic twins complicated by selective intrauterine growth restriction (SIUGR) type III. *J Matern Fetal Neonatal Med*. 2021;34(23):3867-3873.
41. Soni S, Gebb J, Miller K, et al. Predictors of poor outcomes in monochorionic diamniotic twin pregnancies complicated by selective fetal growth restriction. *Fetal Diagn Ther*. 2024;51(3):300-309.

SUPPORTING INFORMATION ON THE INTERNET

The following supporting information may be found in the online version of this article:

Table S1 Collinearity analysis of parameters included in the models.